

题 1. 【简答题】

Given two signal voltage u_1 and u_2 , the corresponding frequency parts are:

$$u_1 = 2\cos 2000\pi t + 0.3\cos 1800\pi t + 0.3\cos 2200\pi t (\text{V})$$

$$u_2 = 0.3\cos 1800\pi t + 0.3\cos 2200\pi t (\text{V})$$

Please answer the following questions:

(1) Are u_1 and u_2 modulated waves? Please list their mathematical expressions.

(2) Find out the power of sideband frequency and the whole power with unit resistor.

(3) Find out the frequency bandwidth B of the modulated wave.

答案:

(1) Analysis u_1

$$(a) u_1 = 2\cos 2000\pi t + 0.3\cos 1800\pi t + 0.3\cos 2200\pi t (\text{V})$$

$$= 2(1 + 0.3\cos 200\pi t)\cos 2000\pi t (\text{V}) \Rightarrow u_1 \text{ is a normal AM wave}$$

(b) $P_{\text{whole}} = P_c + P_{\text{sideband}}$, for a unit resistor

$$P_c = \frac{1}{2} \frac{U_{cm}^2}{R} = 0.5 \times 2^2 = 2 (\text{W})$$

$$P_{\text{sideband}} = \frac{1}{2} m_a^2 P_c = 0.5 \times 0.3^2 \times 2 = 0.09 (\text{W})$$

$$P_{\text{whole}} = P_c + P_{\text{sideband}} = 2.09 (\text{W})$$

$$(c) \text{ frequency bandwidth } B_1 = 2F = 2 \times \frac{\Omega}{2\pi} = \frac{200\pi}{\pi} = 200 (\text{Hz})$$

(2) Analysis u_2

$$(a) u_2 = 0.3\cos 1800\pi t + 0.3\cos 2200\pi t (\text{V}) = 0.6\cos 200\pi t \cos 2000\pi t (\text{V})$$

$\Rightarrow$ so u_2 is a DSB AM wave with suppressed carrier

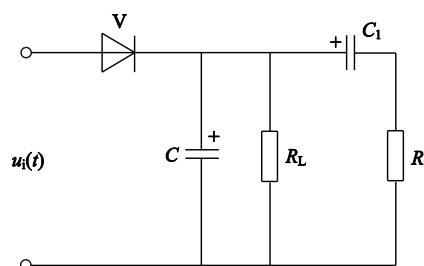
$$(b) P_{\text{whole}} = P_{\text{sideband}} = 0.09 (\text{W})$$

$$(c) \text{ frequency bandwidth } B_2 = 2F = 2 \times \frac{\Omega}{2\pi} = \frac{200\pi}{\pi} = 200 (\text{Hz})$$

题 2. 【简答题】

The demodulation circuit is show in Ex.Fig.4.2 below with

$$u_i(t) = 5\cos 2\pi \times 465 \times 10^3 t + 4\cos 2\pi \times 10^3 t \cos 2\pi \times 465 \times 10^3 t$$



Ex.Fig.4.2

The internal resistance of diode $r_D=100\ \Omega$, $C=0.01\mu\text{F}$ and $C_1=47\mu\text{F}$. In order to ensure no waveform distortion of output signal, please find

- (1) The maximum load resistance R_L of demodulator;
- (2) The minimum input resistance R_i .

答案:

(1) according to the non-distortion condition ,we get

$$R_L < \frac{\sqrt{1-m_a^2}}{m_a} \times \frac{1}{\Omega C} = \frac{0.6}{0.8} \times \frac{1}{2\pi \times 10^2 \times 0.01} = 11.9(\text{k}\Omega)$$

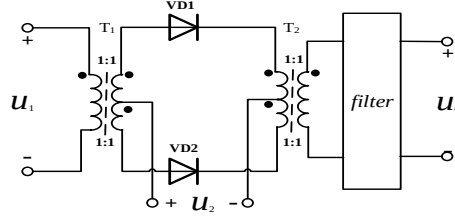
(2) according to the non-clipping distortion condition, we have

$$m_a \leq \frac{R_i}{R_i + R_L}$$

Put $R_L=2.54\text{k}\Omega$ and $m_a=0.8$ into formula above, we get $R_i \geq 47.6\text{k}\Omega$

题 3

The diode multiplier circuit is shown in Ex.Fig.4.3. Assume the I/V curve characteristics of diode VD1 and VD2 are exactly the same, both with slope g_D and intercept 0 (pass the origin), $u_1(t)=U_{1m}\cos\omega_C t$, $u_2(t)=U_{2m}\cos\omega_L t$, $U_{2m} \gg U_{1m}$, $\omega_1=\omega_L-\omega_C$. Ignoring the load effect, please answer



Ex.Fig.4.3

- (1) What type of filter should be used at u_o terminal?
- (2) If the equivalent resistance of filter at the center frequency f_1 is R_L , write the expression of output voltage $u_o(t)$ and list the frequency components;
- (3) Is the circuit able to realize frequency mixing? Why?

(1) Bandpass filter.

$$(2) \quad i_1 = g_D K_1(\omega_L t)(u_1 + u_2), \quad i_2 = g_D K_1(\omega_L t)(-u_1 + u_2)$$

$$\begin{aligned} i_1 - i_2 &= 2g_D K_1(\omega_L t) \cdot u_1 \\ &= 2g_D U_{1m} \cos \omega_c t \left(\frac{1}{2} + \frac{2}{\pi} \cos \omega_L t - \frac{2}{3\pi} \cos 3\omega_L t + \frac{2}{5\pi} \cos 5\omega_L t - \dots \right) \\ u_0(t) = i_1 R_L &= \frac{2}{\pi} g_D U_{1m} R_L \cos(\omega_L - \omega_c)t \\ &= \frac{2}{\pi} g_D U_{1m} R_L \cos \omega_1(t) \end{aligned}$$

(3) Yes, because the intermedium frequency component can be generated in the output voltage u_o .

题 4. 【简答题】

Given the expressions of amplitude-modulated wave, please draw their waveforms and spectrums.

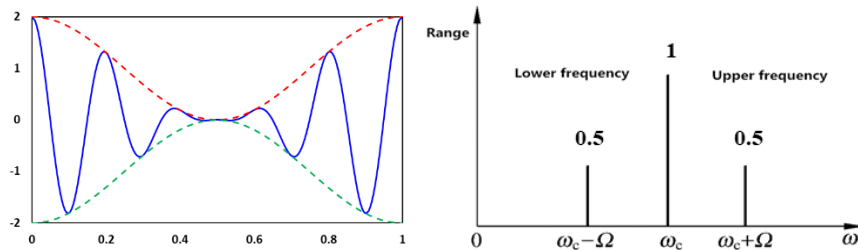
(1) $(1 + \cos \Omega t) \cos \omega_c t$;

(2) $(1 + 0.5 \cos \Omega t) \cos \omega_c t$;

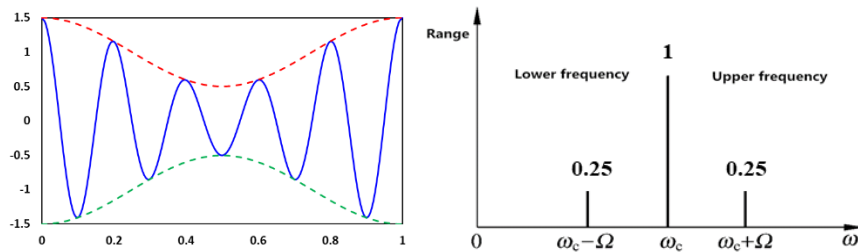
(3) $\cos \Omega t \cos \omega_c t$ ($\omega_c = 5\Omega$).

答案:

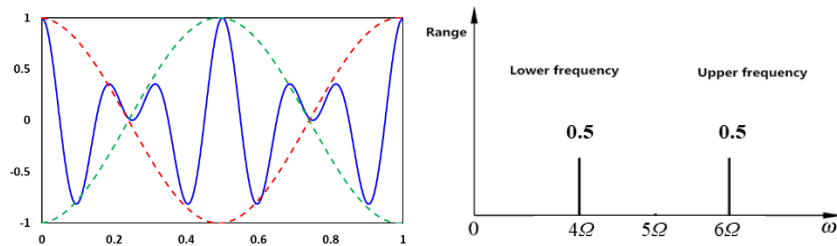
(1)



(2)



(3)



题 5. 【简答题】

In a superheterodyne radio, if intermediate frequency $f_i=f_L-f_c=465\text{kHz}$, tries to analyze what kind of interference does the following phenomenon belong to? And how it was formed.

- (1) When listening to the radio broadcast with frequency of $f_c=931\text{kHz}$, hear a whistle about 1kHz ;
- (2) When listening to the radio station with frequency of $f_c=550\text{kHz}$, hear a strong radio station with frequency of 1480kHz ;
- (3) When listening to the radio station with frequency of $f_c=1480\text{kHz}$, hear a strong radio station with frequency of 740kHz .

答案:

(1) Interference whistle: $2f_c - f_L = 2 \times 931 - 1396 = 466\text{kHz}$

(2) Mirror frequency interference: $f_M = f_c + 2f_i = 550 + 2 \times 465 = 1480\text{kHz}$

(3) Side-channel interference($p=1, q=2$):

$$f_M = \frac{p}{q} f_L \pm \frac{1}{q} f_i = \frac{p}{q} f_c + \frac{p \pm 1}{q} f_i = \frac{1}{2} f_c = 740 \text{ kHz}$$

题 6 【简答题】

The operation frequency range of a heterodyne radio is from $535 - 1605 \text{ kHz}$ with the intermediate frequency of 465kHz . When listening to the radio station at $f_c=700\text{kHz}$, at which other frequencies may we receive the same radio signal(besides f_c)? Please list the two strongest frequencies and explain why that happens.

答案:

(1) $f_L=f_s+f_i=700+465=1165(\text{kHz})$

When $m=n=1, f_{n1}=f_L+f_i=1165+465=1630(\text{kHz})$, which is Mirror Image Interference

(2) When $m=1, n=2, f_{n2}=(mf_L+f_i)/n=(1165+465)/2=815(\text{kHz})$

When $m=1, n=2, f_{n2}=(mf_L-f_i)/n=(1165-465)/2=350(\text{kHz})$, which are Third-channel Interferences